Marksheet✓

Setup & Run Guide

AI answer-sheet evaluation: install on a new Windows laptop and run end to end.

Windows 10 / 11
Python 3.13 · Node.js LTS
FastAPI · React · SQLite · Gemini

~15 minutes	2 installs	2 double-clicks
total, mostly downloads	Python + Node.js, the rest is automatic	`setup.bat` once, then `run.bat`

1 · Python	2 · Node.js	3 · Unzip	4 · API key	5 · setup.bat	6 · run.bat
install	install	project	from Google	one time	every time

1 What you need to install

Install these yourself (one time)

SOFTWARE	VERSION	WHY IT IS NEEDED	DOWNLOAD
Python	3.11 or newer 3.13 recommended	Runs the backend server and the AI pipeline	python.org/downloads
Node.js	20.19 or newer LTS 22 / 24 recommended	Builds the web interface	nodejs.org
Web browser	Chrome, Edge or Firefox	To use the app	Already on Windows
Gemini API key	Free	Reads handwriting, grades answers, compares meaning	aistudio.google.com/apikey

> **Not needed:** Git, a database server, Docker, a GPU or CUDA. The database is a single file and the AI runs in Google's cloud.

Installed automatically by `setup.bat`

PACKAGE	VERSION	PURPOSE
fastapi	0.141.1	Web API framework
uvicorn[standard]	0.54.0	Web server
python-multipart	0.0.32	File uploads
pydantic / pydantic-settings	2.13.5 / 2.15.0	Data validation, reads `.env`
sqlmodel / SQLAlchemy	0.0.47 / 2.0.54	SQLite database access
google-genai	2.25.0	Gemini API client
pymupdf	1.28.2	PDF to page images

opencv-python-headless	5.0.0.93	Image cleanup (straighten, contrast)
pillow	12.3.0	Image loading, phone-photo rotation
numpy	2.5.3	Image arrays
rapidfuzz	3.14.6	Measures agreement of the two handwriting readers
React / Vite / TypeScript (npm)	19 / 8 / 6	Web interface (pinned in `package-lock.json`)

Internet is needed during setup (about 150 MB of downloads) and while evaluating, because every evaluation calls the Gemini API.

2 Install on the new laptop

1 Install Python

1. Download the latest **Python 3.13 Windows installer (64-bit)** from python.org/downloads.
2. Run it. On the **first screen**, tick **"Add python.exe to PATH"**.
3. Click **Install Now** and wait for "Setup was successful".

> **Don't skip:** the "Add python.exe to PATH" box is the most commonly missed step. Without it, setup can't find Python.

2 Install Node.js

1. Download the **LTS** Windows installer (`.msi`) from nodejs.org.
2. Run it and click **Next** through every screen, keeping the defaults. You can leave the "necessary tools / Chocolatey" option unticked.

3 Check both are installed (optional)

Open a **new** PowerShell window (Start → type `powershell` → Enter) and run:

```
python --version     # should print  Python 3.13.x
node --version       # should print  v22.x or v24.x
```

4 Unzip the project

1. Right-click `Marksheet.zip` → **Extract All...** → **Extract**.
2. Pick a simple location such as `C:\Users\<you>\Documents\`. This creates the folder `Ai_Project_Test`.
3. Avoid OneDrive-synced folders if possible; syncing can lock files during install.

5 Get your Gemini API key

Skip this if the zip was made with `-IncludeKey`.

1. Open aistudio.google.com/apikey and sign in with a **personal Gmail** account (company accounts are often blocked).
2. Click **Create API key** and copy the key.

> **"Unable to create API key"?** Go to console.cloud.google.com/projectcreate, create a project (name `answer-sheet-eval`, parent "No organization"), come back and choose **Create API key in existing project**.

6 Double-click `setup.bat`

1. Open the `Ai_Project_Test` folder and **double-click** `setup.bat`.
2. If a blue **"Windows protected your PC"** box appears: click **More info** → **Run anyway**.
3. When asked, **paste your API key** and press `Enter`.
4. Wait for **"Setup complete"** (2–5 minutes), then press any key.

What you will see:

```
==> Checking Python                          Python 3.13.x OK
==> Checking Node.js                         Node.js v24.x OK
==> Setting up the Python environment        (creates backend\.venv, installs packages)
==> Installing frontend packages (npm)
==> Building the web interface
==> Configuring the Gemini API key           Paste your Gemini API key: ********
==> Testing the installation                 Backend imports OK
                                             Gemini key OK (61 models available)
Setup complete. Start the app by double-clicking run.bat
```

> **Safe to repeat:** `setup.bat` can be run again at any time. It repairs or skips whatever is already done.

3 Run the app

▶ Double-click `run.bat`

1. A black window opens. After about 3 seconds your browser opens **http://localhost:8000**.
2. **Keep the black window open** while you use the app.
3. To stop: close the black window (or press `Ctrl`+`C` in it).

From now on this is the **only step** needed each time you want to use the app.

Using the app

#	WHERE	WHAT TO DO
1	**Evaluate** page	Drag the answer-sheet pages (JPG, PNG or PDF) into the drop box and put them in page order with ‹ ›.

2	Rubric	Choose the answer key, e.g. *Python Programming — Core Concepts*.
3	**Evaluate** button	Progress runs through Extract, Structure, Score and Confidence. The first run of a new sheet takes about 20–60 s; re-running the same sheet takes about 1 s.
4	Results	For each question: the handwriting as read, ✓ / ½ / ✗ per mark point with evidence, the score, and a confidence label with its reason. Click **Why?** for all signals.
5	Filters and export	**Needs review** shows answers a human should check. **CSV** / **JSON** download the results.
6	**Rubrics** page	View, edit or create answer keys (form or JSON).

Other ways to run (from the project folder)

GOAL	COMMAND
Start without `run.bat`	`powershell -ExecutionPolicy Bypass -File .\run.ps1`
Developer mode (page reloads as you edit code)	`powershell -ExecutionPolicy Bypass -File .\run.ps1 -Dev` → localhost:5173
Command line only (writes `outputs\results.json` + `.csv`)	`cd backend` `.\.venv\Scripts\python.exe cli.py ..\data\samples`
Backend server by hand	`cd backend` `.\.venv\Scripts\python.exe -m uvicorn app.main:app --port 8000`
Interactive API docs	Open localhost:8000/docs while the app is running

4 Project layout and where data is stored

```
Ai_Project_Test/
├── setup.bat / setup.ps1        one-time installer (new laptop)
├── run.bat   / run.ps1          start the app
├── make-zip.ps1                 make a clean, portable zip
├── SETUP.md · README.md · APPROACH.md
├── backend/
│   ├── .env                     Gemini key + model settings   (private: don't share)
│   ├── .env.example             template for .env
│   ├── requirements.txt         Python packages (exact versions)
│   ├── cli.py                   command-line runner
│   ├── app/                     FastAPI app + pipeline: extract → segment → score → confidence
│   └── storage/                 ALL app data
│       ├── app.db               SQLite database: rubrics + evaluation history and results
│       ├── cache.db             Gemini response cache (safe to delete)
│       └── evals/<id>/          cleaned page images for each evaluation
├── frontend/                    React + Vite web interface  (src = code, dist = built site)
├── data/
│   ├── QUESTION_PAPER.md        questions + short answer key
│   ├── SOURCE.md                source of the sample sheet
│   ├── rubrics/                 rubric JSON (loaded into the database on first start)
│   └── samples/                 handwritten answer pages
└── outputs/                     results.json / results.csv
```

Settings in `backend\.env`

SETTING	MEANING
`GEMINI_API_KEY`	Your key (required)
`PRIMARY_MODELS`	Handwriting reader and grader, tried left to right; the next model is used automatically when one is busy
`CROSSCHECK_MODELS`	Second, independent handwriting reader; how well it agrees with the first feeds the confidence label
`EMBED_MODEL`	Meaning-similarity sanity check (default `gemini-embedding-2`)
`JUDGE_RUNS`	Grading runs per sheet (default 3); disagreement between runs lowers confidence

5 Troubleshooting

PROBLEM	FIX
`python` or `node` "is not recognized"	Close **all** terminals and open a new one, or restart the laptop. For Python: re-run the installer → **Modify** → tick "Add Python to environment variables".
`python` opens the Microsoft Store	Settings → Apps → Advanced app settings → **App execution aliases** → turn off both "python" entries.
"running scripts is disabled on this system"	Use the `.bat` files, or run once: `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`

`[Errno 10048]` only one usage of each socket address	The app is already running on port 8000. Use the open app, or close the other black window first.
Browser shows `ERR_CONNECTION_REFUSED`	The server isn't running. Start `run.bat` and keep its window open.
Evaluation fails with **503** / "high demand"	Google's free tier is overloaded. Wait a minute and try again; pages already read are cached.
429 / quota / rate limit	Free-tier limit reached. Wait a minute (or until the next day for the daily limit).
404 "model … no longer available"	Google retired a model. Update `PRIMARY_MODELS` / `CROSSCHECK_MODELS` in `backend\.env` to current names.
"API key not valid" / 400 / 403	Put a fresh key in `backend\.env` as `GEMINI_API_KEY=...` (no quotes or spaces) and restart `run.bat`.
`pip install` or `npm ci` fails	Check the internet connection, company proxy or antivirus, then re-run `setup.bat`.
Page looks old after editing code	`cd frontend` → `npm run build`, then restart `run.bat`.
Start fresh (erase history)	Stop the app, delete `backend\storage\`, start again. The rubric reloads from `data\rubrics`.

6 Security checklist

- `backend\.env` holds your API key. Don't share it, upload it to GitHub, or send a zip made with `-IncludeKey`.
- If the key was ever pasted anywhere public (chat, email, screenshot), delete it at aistudio.google.com/apikey and create a new one.
- On Gemini's free tier, Google may use inputs to improve its products. Use sample scripts, not real students' papers.
- The app listens on `localhost` only, so other computers on the network can't reach it.

Appendix: moving the project again

Don't zip the folder by hand. `backend\.venv` and `frontend\node_modules` (about 420 MB) are tied to one PC. From the project folder, run:

```
powershell -ExecutionPolicy Bypass -File .\make-zip.ps1          # key left out (recommended)
powershell -ExecutionPolicy Bypass -File .\make-zip.ps1 -IncludeKey  # key included
```

This creates `Marksheet.zip` (about 1 MB) with the code, rubrics, samples, outputs and database. On the other laptop, follow section 2 again.